

Chapter 1: Directed Numbers**Chapter 2: Squares, Square Roots, Cubes and Cube Roots****Chapter 3: Algebraic Expression II****Chapter 4: Linear Equations I****Part A: Objective Questions [20 marks]**

Instruction: Answer all questions. Each question is followed by four choices of answers. Choose the correct answer. Show the working steps at the space provided.

1. Which of the following calculation has the smallest value?

A. $(-2) + (-8)$
B. $0 - 10$
C. $(-18) + (-20)$
D. $34 + (-29)$

2. Diagram below shows a sequence of numbers.

$-8, p, 0, 4, q, 12$

Find the value of $p - q$.

A. 12
B. -12
C. 4
D. -4

3. $56 + (40 - 88) \times 10 =$

A. 80
B. 116
C. -424
D. -784

4. $2\frac{1}{4} \times \left(\frac{1}{2} + \frac{5}{7}\right) =$

A. $1\frac{47}{56}$
B. $1\frac{27}{56}$
C. $2\frac{17}{28}$
D. $2\frac{41}{56}$

5. $\frac{-35 + (-14)}{5 - (-2)} =$
- A. -7
B. 7
C. -14
D. 14
6. Which of the following is true?
- A. $\left(\frac{3}{4}\right)^2 = \frac{6}{8}$
B. $\left(\frac{3}{4}\right)^2 = \frac{6}{16}$
C. $\left(\frac{4}{7}\right)^2 = \frac{8}{14}$
D. $\left(\frac{4}{7}\right)^2 = \frac{16}{49}$
7. Which of the following is not a perfect square?
- A. 225
B. 256
C. 47
D. 49
8. 3.45^2 is between
- A. 3 and 4
B. 11 and 12
C. 16 and 17
D. 19 and 20
9. Given $\sqrt{45.1256} = 6.72$ and $\sqrt{451.256} = 21.24$. Find the value of $\sqrt{4512.56}$.
- A. 6.72
B. 67.2
C. 21.24
D. 212.4
10. A woman walks 5 rounds along the border of a square field of area 49 m^2 . Find the total distance she walks in m.
- A. 7
B. 28
C. 49
D. 140

11. Identify the unknowns in the algebraic term $20x^3y^2$.

- A. x only
- B. y only
- C. x and y
- D. x^3y^2

12. $2(3x + 2y) + (5x - 3y) =$

- A. $11x + y$
- B. $11x + 7y$
- C. $10x + 8y$
- D. $10x - 10y$

13. Given that $x = 5$ and $y = -2$, find the value of

$$2x^2 - xy.$$

- A. 10
- B. 40
- C. 50
- D. 60

14. Simplify the following expression.

$$\frac{12x^2y^3 \times 8z}{4y^2z}$$

- A. $6x^2y$
- B. $6x^2y^5z^2$
- C. $24x^2y$
- D. $24x^2y^5z^2$

15. Simon bought x pens for RM 6 each and y books for RM 2 each. She sells the pens for RM 8 per pen and the books for RM 5 per book. Which of the following expressions shows the total profit, in RM, for the sale of pens and books?

- A. $2x + 3y$
- B. $2x + 7y$
- C. $3x + 2y$
- D. $15x + 7y$

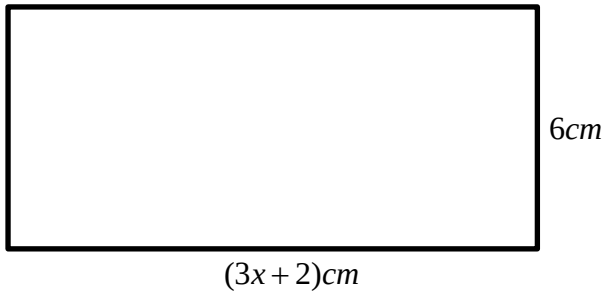
16. Given $25x + y = -14$. Find the value of y .

- A. 39
- B. -39
- C. 11
- D. -11

17. Given $\frac{x-5}{7} = 3$, therefore, $x =$

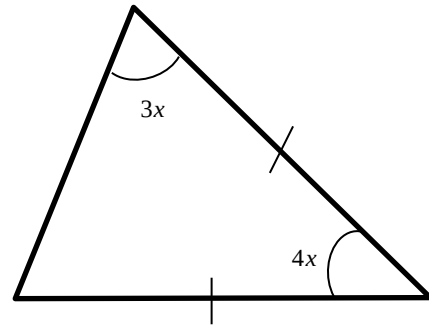
- A. 15
- B. 21
- C. 26
- D. 30

18. The area of the rectangle below is 66 cm^2 . Find the value of x .



- A. 3
- B. 5
- C. 7
- D. 9

19. Diagram below shows an isosceles triangle.



What is the value of x ?

- A. 18
- B. 26
- C. 28
- D. 36

20. If 0.3 of x is RM 57, calculate the value of x .

- A. RM 169
- B. RM 171
- C. RM 190
- D. RM 399

Part B: Subjective Questions [40 marks]

Instruction: Answer all questions. Show ALL the working steps at the space provided. The usage of calculators is NOT allowed.

1. (a) Find the value of:

$$-49 + 32 \div 1\frac{3}{5}$$

[1 mark]

- (b) Calculate the value of
 $184 \div 2(15 + 7 - 18)$

[2 marks]

2. Calculate the value of $12 - \frac{1}{6} \times 2.85$.
Express the answer correct to two decimal places.

[3 marks]

3. Solve each of the following. Express all the answer correct to three decimal places.

(a) 0.52^2

[1 mark]

(b) $\frac{2.5^2}{3^3}$

[2 marks]

4. (a) Given that $1.7^2 = 2.89$, find the value of $\sqrt{0.0289}$.

[2 marks]

(b) A box in the shape of cube has base area of 144 cm^2 . Calculate its volume, in cm^3 .

[3 marks]

5. Simplify each of the following expressions:

(a) $(15 - \frac{20y}{2}) \div 5$

[2 mark]

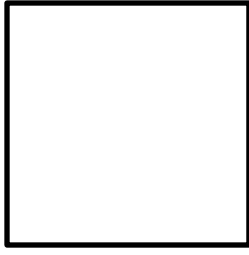
(b) $\frac{2}{7}(7x - 21y) - 4(-x - 2y)$

[2 marks]

6. (a) Simplify $\frac{2x-1}{3} - \frac{x-2}{4}$.

[2 marks]

- (b) The diagram below shows a square.



$$(9x + 6y - 3)cm$$

Calculate the perimeter of the square.

[2 marks]

7. (a) Alex is x years younger than his brother. If his brother is $3y$ years old in 2014, write an expression for Alex's age in 2020.

[2 marks]

- (b) Simplify
 $12ab^3 + 4b - [ab - a(b - ab^2) + 3b]$

[3 marks]

8. Solve each of the following linear equations:

(a) $\frac{x}{2} = -5$

[1 mark]

(b) $3(x - 4) = x + 2$

[2 marks]

9. (a) Solve $\frac{3x - 6}{4} = \frac{6x + 7}{5}$

[3 marks]

- (b) A bag contains $(2p - 5)$ red balls and $(18 + 3p)$ yellow balls. If the number of red balls is equivalent to $\frac{1}{3}$ of the yellow balls, find the value of p .

[3 marks]

10. The length of the rectangle is $(q + 7)$ cm while its breadth is q cm. If the perimeter of the rectangle is 56 cm, what is the value of q ? What is the area of the rectangle?

[4 marks]